Fiscal Year:	FY 2021	**Task Last Updated:**	FY 01/19/2023
PI Name:	Bowles, Dawn Ph.D.		
Project Title:	Gene Therapy Countermeasures for Detrimental Effects of Space Radiation		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	TRISH--TRISH		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	27710-0001	**Congressional District:**	4
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2018 TRA BRASH1801: Translational Research Institute for Space Health (TRISH) Biomedical Research Advances for Space Health
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No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	TRISH
Contact Monitor:		**Contact Phone:**	
Contact Email:			
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Key Personnel Changes/Previous PI:			
COI Name (Institution):	Bishawi, Muath M.D. (Duke University)		
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Task Description:	The risk of exposure to space radiation on astronaut health is not expected to be eliminated through current approaches and requires innovation. Gene therapy has been developed as a methodology to introduce genetic information into cells to compensate for abnormal genes or to make a beneficial, protective protein. Gene delivery vehicles or vectors based upon the non-pathogenic adeno-associated virus (AAV) confer sustained expression (up to years) of therapeutic nucleic acids as demonstrated in numerous preclinical and clinical trials in human diseases such as muscular dystrophy, cystic fibrosis, Alzheimer's, and Parkinson's. Already 2018 has been a banner year for gene therapy with the US Food and Drug Administration's first ever approval of an AAV-based treatment, Luxturna, which essentially cures an inherited form of blindness. Excitement over the future use of AAV-based gene therapy to cure much of humankind's ailments is evidenced by recent investments into this technology by large pharma (such as Pfizer and Novartis) to escalate improvements in vector technology, identify new therapeutic targets, and to scale up and lower production costs. One future application of gene therapy is for NASA to use it as a prophylactic to protect astronauts from stressors of space exploration prior to initiation of travel. The use of an adeno-associated viral vector based strategy prophylactically avoids the need to store a pharmaceutical agent during exploration missions. The development of successful gene therapy strategies relies upon four experimental components: 1) a well-defined disease or injury state and disease models, 2) molecular targets appropriately therapeutic for the disease, 3) a vector (usually viral based), and 4) a delivery approach. The overall objective of this proposal is to perform proof of concept studies addressing these four components in the development of a gene therapy to improve an astronaut's resilience against space radiation. This project, if successful, may lead to the development of a novel gene therapy based approached that could be administered safely to the astronauts prior to space exploration mission. Furthermore, as AAV-based gene delivery is known to have lasting effects without integration of nucleic acid in the genome, an AAV-based treatment may lead to long-term expression of therapeutic nucleic acid in the astronaut's tissues and organs up to years, a time line consistent with the projected length of the space exploration missions.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Gene therapy is an advanced treatment approach that alters the genetic composition of cells to confer therapeutic protein or RNA expression to the target organ. This methodology introduces genetic information into cells to compensate for abnormal genes, to make a beneficial, protective protein, or to alter pathways in a beneficial manner. Gene delivery vehicles or vectors based upon the non-pathogenic adeno-associated virus (AAV) confer sustained expression (up to years) of therapeutic nucleic acids as demonstrated in numerous preclinical and clinical trials in human diseases as well as U.S. Food and Drug Administration (FDA) approved treatments for inherited blindness and spinal muscular atrophy. Excitement over the future use of AAV-based gene therapy to cure much of mankind's aliments is evidenced by recent investments into this technology by large pharma (such as Pfizer, Bayer, Novartis) to escalate improvements in vector technology, identify new therapeutic targets, and to scale up and lower production costs. One future application of gene therapy is for NASA to use it as a prophylactic to protect astronauts from stressors of space exploration prior to initiation of travel. The use of an AAV vector based strategy prophylactically avoids the need to store a pharmaceutical agent during exploration missions. This project, if successful may lead to the development of a novel gene therapy based approached that could be administered safely to the astronauts prior to space exploration mission. Furthermore, as AAV-based gene delivery is known to have lasting effects without integration of nucleic acid in the genome, an AAV-based treatment may lead to long-term expression of therapeutic nucleic acid in the astronaut's tissues and organs up to years, a time line consistent with the projected length of the space exploration missions. The innovative technologies developed are AAV-SPRR1a and AAV-dsup vectors.
Task Progress:	Original project aims/objectives: The risk of exposure to space radiation on astronaut health is not expected to be eliminated through current approaches and requires innovation. Gene therapy has been developed as a methodology to introduce genetic information into cells to compensate for abnormal genes or to make a beneficial, protective protein. Gene delivery vehicles or vectors based upon the non-pathogenic adeno-associated virus (AAV) confer sustained expression (up to years) of therapeutic nucleic acids. One future application of gene therapy is for NASA to use it as a prophylactic to protect astronauts from stressors of space exploration prior to initiation of travel. The use of a recombinant AAV (rAAV) based strategy prophylactically avoids the need to store a pharmaceutical agent during exploration missions. The use of an adeno-associated viral vector based strategy prophylactically avoids the need to store a pharmaceutical agent during exploration missions. The development of successful gene therapy strategies relies upon four experimental components: 1) a well-defined disease or injury state and disease models, 2) molecular targets appropriately therapeutic for the disease, 3) a vector (usually viral based), and 4) a delivery approach. The overall objective of this proposal is to perform proof of concept studies addressing these four components in the development of a gene therapy to improve an astronaut's resilience against space radiation in the following specific aims: Aim1) Using an established rodent tissue repository, explore expression of potential therapeutic targets hypothesized to confer radiation protection (hspa5, hspa9, AIFM1, SPRR1a); Aim 2) Evaluate abilities of identified targets (hspa5, hspa9, AIFM1, SPRR1a, tardigrade dsup) to confer radiation protection using stable 293 based cell lines or cells transduced with rAAV overexpressing these targets; Aim 3) Evaluate the ability of down-selected rAAVs to confer radiation protection in mouse models. Key Findings: Aim 1: Using a rodent tissue repository established by our lab, western blotting was performed on homogenized heart apexes (150 cGy Galactic Cosmic Ray (GCR)5-ion and age matched sham irradiated controls) using SPRR1a, AIFM1, and hspa5 primary antibodies. Western blot analysis of SPRR1a, AIFM1, and hspa5 of the hearts of GCR and control mice 24 hours post radiation demonstrated a significant increase in the level of in band2 of sprr1a post irradiation but no other changes in proteins levels of the tested proteins. However, in contrast, at 12 months post radiation, there were significant decreases in the levels of most of the SPRR1a bands, as well as trending decreases in the levels of AIFM1 and hspa5. Aim 2: 293 stable cell lines and rAAVs containing the coding sequences of hspa5, hspa9, AIFM1, SPRR1a, and the tardigrade dsup were generated. This was accomplished by purchase of coding sequence, PCR amplification, and cloning into a vector utilized for the generation of rAAV. In addition, the plasmids generated also contained neomycin expression cassette sequence, so these constructs could also be utilized in the generation of stable cell lines following transfection into 293 cells and G418 selection. Cells overexpressing SPRR1a, hspa9, or dsup exhibited advantages over control cell lines in a variety of cell based assays post radiation (cell growth, caspase 3/7, LDH). Rat neonatal cardiomyocytes (RNNCs) or A549 cells were transduced with rAAV (dsup, Hspa5, Hspa9, SPRR1a, AIFM1, or firefly

	luciferase: 80,000 genome copies/cell) for 48hrs before radiation. Benefit of several of these vectors was seen in A549 cells but not in RNNCs. We are still evaluating the results of AAV vector experiments in both A549 and RNNCs. Aim 3: C57Bl6 male mice (Jackson Laboratory) were injected at approximately 15 weeks of age with 1 x 1011 viral particles of rAAV (dsup, SPRR1a, or Firefly luciferase) either intravenous (IV), intraperitoneal (IP) or via direct cardiac injection (DCI). All mice were transported to Brookhaven National Lab (BNL) for 150 cGy GCR5-ion irradiation at 21 weeks of age (approximately 6 weeks post vector delivery) and returned to Duke University for longitudinal studies. At ~5 months post-radiation, DCI mice were evaluated by echocardiography (n= 50) and non-survival hemodynamic Pressure-Volume Loop (PVL) assessment (n= 49). Luc rAAV injected mice were imaged using IVIS Kinetic (Perkin Elmer) to validate luciferase gene expression (n= 39). Some mice received pulmonary function tests (PFTs; n=29) to assess any changes to the pulmonary system post-radiation and rAAV treatment. All mice were euthanized at approximately 5 months post-radiation and tissues bio banked, via LN2 flash freezing. We are still analyzing echocardiographic, hemodynamic, PFT, and histology of mice. Future Direction & Conclusions: This project experienced many delays and was significantly impacted by COVID. In addition to the laboratory shutdown period at Duke, the project was delayed due to laboratory social distancing requirements, availability of protective equipment (PPE), and project supplies. Most significantly affected was Specific Aim 3 that evaluated the down selected rAAV in mice. Because of these delays, the mice were only survived 5 months post radiation instead of the 9-12 months that we originally proposed. This would have provided a longer time to see the development of cardiac dysfunction as we reported in Bishawi et al 2022. Regardless, this funding allowed us to demonstrate proof of concept of gene therapy based radiation protection in cell based models as well as feasibility of administration in animal models. Importantly, we demonstrated long lasting gene expression at 5 months post radiation in the mice. Overall continuation of this line of research may lead to the development of a novel gene therapy based approached that could be administered safely to the astronauts prior to space exploration mission.
Bibliography Type:	Description: (Last Updated: 06/30/2026)
Articles in Peer-reviewed Journals	Davis CM, Allen AR, Bowles DE. "Consequences of space radiation on the brain and cardiovascular system." J Environ Sci Health C Toxicol Carcinog. 2021;39(2):180-218. https://doi.org/10.1080/26896583.2021.1891825 · PMID: 33902387 , Apr-2021
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